You need to:

- **Use current units of mass, length, area, volume and capacity in practical situations and express quantities in terms of larger or smaller units.**

Recap

Units of length

10 mm = 1 cm

100 cm = 1 m

1000 m = 1 km

Units of mass

1000 g = 1 kg

1000 kg = 1 tonne

Units of capacity

1 cm^3 = 1 ml

1000 ml = 1 litre

Units of area

100 mm^2 = 1 cm^2

10 000 cm^2 = 1 m^2

1 000 000 m^2 = 1 km^2

Units of volume

1000 mm^3 = 1 cm^3

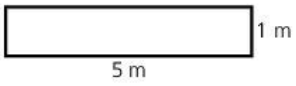
1 000 000 cm^3 = 1 m^3

Watch out!

Units of area and units of volume have different conversions to the standard units of length. It is useful to **learn** these conversions and remember to use them in the exam.

Worked example

Calculate the number of square centimetres in 5 square metres. **[2 marks]**



1 m × 5 m = 100 cm × 500 cm

= 50 000 cm^2

Exam tip

Imagine a rectangle with dimensions such that the area is 5 square metres.

Alternatively, use your learnt conversion from m^2 to cm^2 so 5 × 10 000 = 50 000.

Questions

1 Complete these unit conversions.

a 1 m^2 = ... cm^2

b 1 km^2 = ... m^2

c 20 000 cm^2 = ... m^2

d 50 000 m^2 = ... km^2

e 14.8 km^2 = ... m^2

f 1 380 000 cm^2 = ... m^2

g 0.25 m^2 = ... cm^2

h 0.00436 km^2 = ... m^2

i 0.000 000 547 km^2 = ... mm^2

2 Complete these unit conversions.

a 1 cm^3 = ... mm^3

b 1 m^3 = ... cm^3

c 200 000 cm^3 = ... m^3

d 6500 mm^3 = ... cm^3

e 1000 cm^3 = ... litres

f 1 m^3 = ... litres

g 6.7 litres = ... cm^3

h 1 km^3 = ... m^3

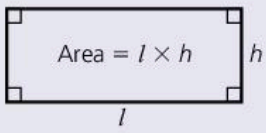
5.2 Mensuration

You need to:

- **Carry out calculations involving the perimeter and area of a rectangle, triangle, parallelogram and trapezium, and compound shapes derived from these.**

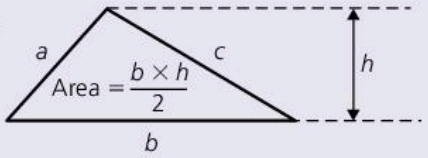
Recap

Rectangle



Perimeter = $2l + 2h$

Triangle

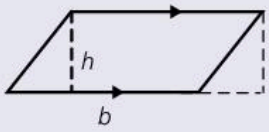


Perimeter = $a + b + c$

Parallelogram

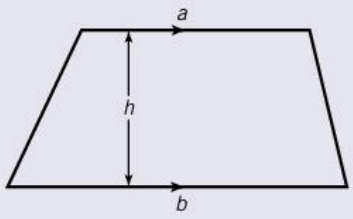
The area of the parallelogram is the same as the area of the dotted rectangle.

So area of parallelogram = $b \times h$, that is base × height



Trapezium

The area of a trapezium is $\frac{1}{2}(a+b) \times h$



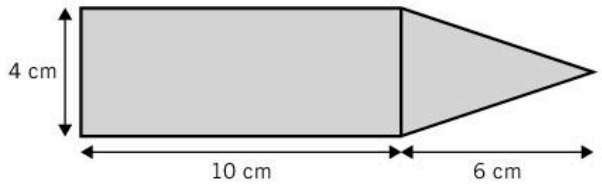
Key skills

You need to be able to use the above formulae to find the area and/or perimeter of shapes made up from the above shapes. These are called **compound** shapes.

Exam tip

Split the calculation into two parts and make sure your working is set out clearly.

Worked example



Calculate the area of this shape. **[3 marks]**

Area of rectangle = $4 \times 10 = 40$ cm^2

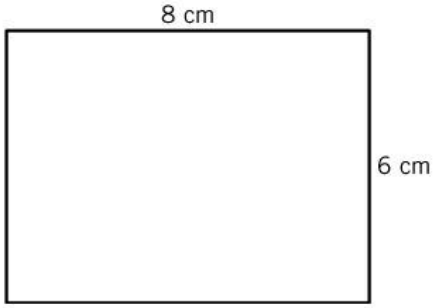
Area of triangle = $\frac{4 \times 6}{2} = 12$ cm^2

Total area = $40 + 12 = 52$ cm^2

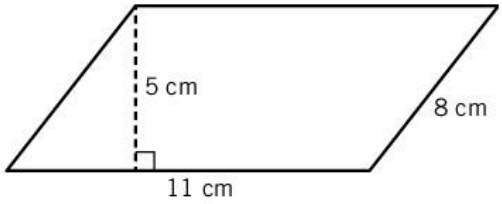
Questions

1 Find the area and perimeter of these shapes.

a

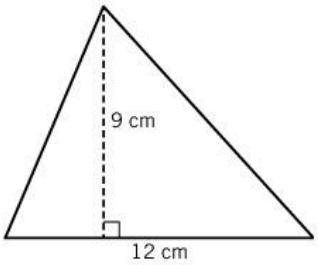


b

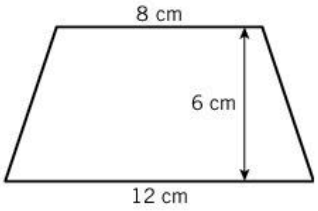


2 Find the area of these shapes.

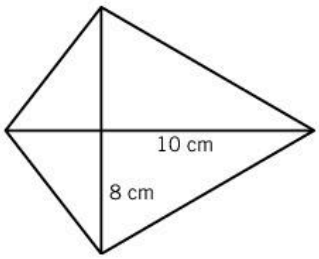
a



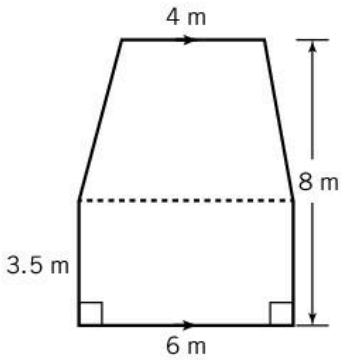
b



c



3 Find the area of this compound shape.



4 A rectangle has area 126 cm^2 and width 4.5 cm. Calculate:

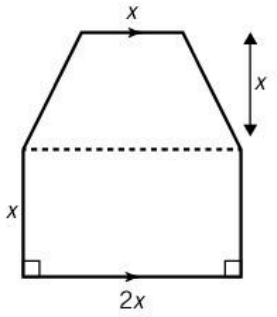
a the length of the rectangle

b the perimeter of the rectangle.

5 A triangle has area 52 cm^2 and base 2.6 cm. Calculate the height of the triangle.

6 A trapezium of area 210 cm^2 has parallel sides of length 10 cm and 18 cm. How far apart are the parallel sides?

7 The area of this compound shape is 56 cm^2.



Find the value of x.

To **Raise your grade** now try question 7, page 143

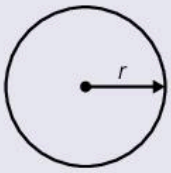
5.3 Mensuration

You need to:

- Carry out calculations involving the circumference and area of a circle.
- Solve problems involving the arc length and sector area as fractions of the circumference and area of a circle.

Recap

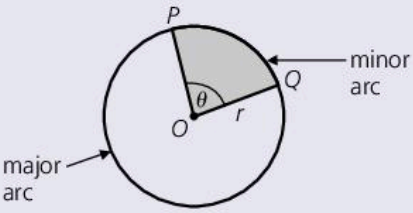
Circle



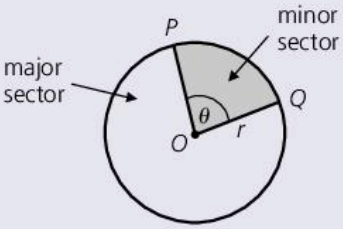
Area of circle $= \pi r^2$

Circumference of circle $= 2\pi r$

Sector of circle



Length of minor arc $= \frac{\theta}{360} \times 2\pi r$



Area of minor sector $= \frac{\theta}{360} \times \pi r^2$

Watch out!

You may be asked for the answer as a rounded decimal or as a multiple of π.

If it doesn't specify, it is best to give both.

Key skills

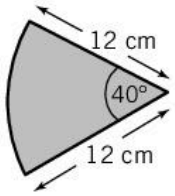
You need to be able to use the formulae for the length of an arc and the area of a sector.

Worked example

A large circular cheese has radius 12 cm.
A wedge is cut from the cheese in the shape of a sector of angle 40°.
Calculate:

(a) the cross-sectional area of the wedge

(b) the total perimeter of the wedge. **[5 marks]**



(a) Area of sector $= \frac{40}{360} \times \pi \times 12^2 = 16\pi = 50.3$ cm^2 (to 3 s.f.)

(b) Length of arc $= \frac{40}{360} \times 2 \times \pi \times 12 = \frac{8}{3}\pi = 8.38$ cm (to 3 s.f.)

Perimeter $= \frac{8}{3}\pi + 2 \times 12 = 32.4$ cm (to 3 s.f.)

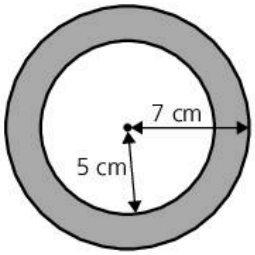
Exam tip

Don't forget to add on the two radii for the **total** perimeter.

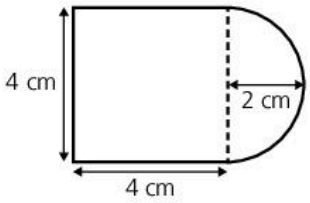
Use the exact answer for the arc length to ensure no rounding errors are carried forwards.

Questions

1 Find, in terms of π, the area which is enclosed between two circles with the same centre, one with radius 7 cm and the other with radius 5 cm.



2 Show that the area of a shape consisting of a square of side length 4 cm with a semicircle of radius 2 cm added to one side is $16 + 2\pi$.



3 Find the radius of the circle which has the same area as the combined area of a circle of radius 12 cm and a circle of radius 5 cm.

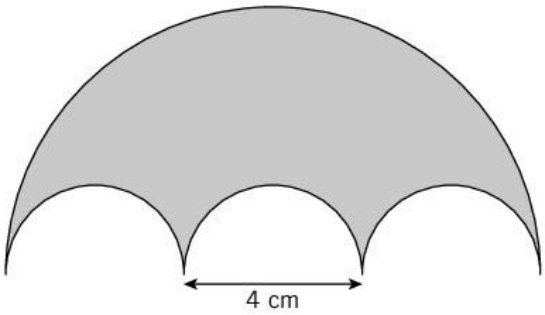
4 A running track is 60 metres wide. The inside of the inner lane consists of two straight sections of length 80 m and two semicircles of radius 24 m. The outside of the outer lane consists of two straight sections of length 80 m and two semicircles of radius 30 m.

a Find, in terms of π, the perimeter of the inside of the inner lane.

b Find, in terms of π, the perimeter of the outside of the outer lane.

c Find, in terms of π, the area of the track.

5 Calculate the area of this shape. Each of the three smaller semicricles has diameter 4 cm. Leave your answer in terms of π.

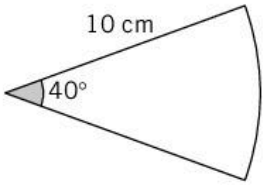


6 A circle has an area of 100 cm^2.
What is its radius? Give your answer correct to 3 significant figures.

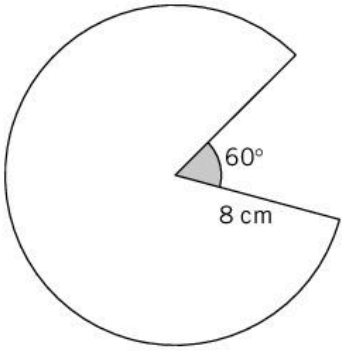
7 A circle has a circumference of 50 m.
What is its area? Give your answer correct to 3 significant figures.

8 Find the area and perimeter of these sectors, correct to 3 significant figures.

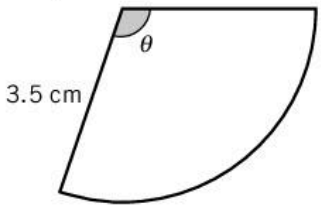
a



b



9 The perimeter of this sector is 14.33 cm.



Find the value of θ.

To **Raise your grade** now try questions 2, 5 and 6, pages 142–143